YieldFlo User Manual

YieldFlo is a yield monitoring application that works alongside AgOpenGPS (AOG). It records yield rate, moisture, and GPS position continuously while harvesting, and produces spatial yield maps and job reports.

YieldFlo receives GPS and section control data from AOG over a local network (UDP). A YieldFlo hardware module connects to the combine's clean-grain elevator and moisture sensor and sends live readings to the PC application.

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1. Getting Started

Requirements

Item	Requirement
Operating system	Windows 10 or Windows 11
.NET Framework	4.8 (pre-installed on Windows 10 May 2019 Update and later)
AgOpenGPS	Running on the same PC or local network

No installer is required. Extract the YieldFlo folder to any location and run `YieldFlo.exe`.

Hardware module

- YieldFlo module (based on ESP32) mounted on the combine
- Connected to the clean-grain elevator optical sensor
- Connected to the capacitance moisture sensor (optional)
- Communicates with the PC via WiFi or CAN bus

Network

- **WiFi mode:** by default the tablet runs its own hotspot and the module joins it — see Comm. The PC and module can also share an existing WiFi network.
- AOG must be running and broadcasting GPS/section data on the local network

Recommended first-run setup

On first launch YieldFlo creates a default crop, header, and profile. Before starting your first job:

1. Open **Menu** → **Settings** and select your preferred units (bu/ac or t/ha)
2. Power the module and open **Menu** → **Comm**. Answer **Yes** to any questions it asks, and wait for **Module: Connected**
3. Open **Menu** → **Profiles** and enter your combine ID
4. Open **Menu** → **Crops** and add or edit crops for your operation
5. Open **Menu** → **Headers** and enter the correct cutting width
6. Open **Menu** → **Fields** and add field names if desired

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2. Main Screen

The main screen displays live harvest data and job status.

Toolbar buttons

Left to right, as they appear on screen:

Button	Action
Exit	Close YieldFlo
—	Switch to compact (mini) view — see below
Menu	Open the main menu
Start	Start a new job (opens Jobs menu) or resume the paused active job
Pause	Manually pause recording
Stop	Stop and close the current job (requires confirmation)

The menu screen has a **?** button in its title bar that opens this manual.

Data panels

Panel	Description
YIELD	Instantaneous yield rate (bu/ac or t/ha), smoothed over recent readings
MOISTURE	Live grain moisture percentage from the capacitance sensor
Totals area	Total area harvested, total mass, and average yield for the current job

Sensor bars

Two horizontal bar gauges below the data panels:

- **Elev Flow:** Grain obstruction percentage in the clean-grain elevator. Higher = more grain flowing.
- **Moisture:** Relative moisture sensor reading scaled to a 0–30% range.

Status bar

Indicator	Green	Orange	Red / Silver
GPS	AOG connected and sending position	—	No AOG data
Module	Module data received within 5 s	Module connected but the sensor is not reading — shows `MODULE - NO SENSOR`	No module data
Job name	Job recording (active)	Job paused	No active job (silver)

When YieldFlo displays a notification (e.g. export complete, error), a full-width message overlays the status bar for 10 seconds then clears automatically. Yellow text = informational; red text = error.

Compact (mini) view

Press ─ in the top-right corner of the main screen to switch to a small floating overlay that shows only the live yield reading. This is useful when running YieldFlo alongside AgOpenGPS on a single monitor.

The compact overlay can be dragged anywhere on screen. Press the restore button on the overlay to return to the full main screen. Its position is remembered between sessions.

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3. Jobs

Menu: Menu → Jobs

A job records yield, moisture, GPS position, and area for a single harvest session. Each job is linked to a crop, header, and profile.

Starting a job

1. Press **Menu** on the main screen, then **Jobs**
2. Press **New** — a draft job is prepared with a default name and the current Crop/Header/Profile, and the **Field** drop-down is focused
3. Select a **Field** (optional) — this replaces the default job name with the field name; the **Job Name** can still be edited afterward
4. Adjust **Crop**, **Header**, and **Profile** if needed
5. Enter any **Notes** for the job (optional)
6. Press **Save** — this creates the job and starts it immediately (prompts for confirmation if another job is currently active)

Recording begins automatically when harvesting conditions are met (speed > 0.5 km/h and AOG sections are on).

Managing jobs

The jobs list shows all saved jobs with their status, date, area, and linked field. Click a column header to sort by that column.

Action	Description
Load	Make a saved job the active job (prompts if another job is already active)
Delete	Permanently remove a job and all its recorded data — cannot be undone

Only one job can be active at a time.

Auto-pause

The job automatically pauses when harvesting stops:

- Machine speed falls below 0.5 km/h, **or**
- AOG turns all header sections off (e.g. turning on headland, travelling over harvested ground)

When auto-paused the job name turns orange in the status bar. Recording resumes automatically when harvesting conditions return.

Sensor fault pause

Recording also stops if the yield sensor stops reading — the module reports a sensor error, module data stops arriving, or the sensor reads nothing at all for 30 seconds while harvesting. The Module indicator turns orange and shows `NO SENSOR`, and a red message appears if this interrupts a pass.

Clean the sensor lens and check its wiring. Recording resumes automatically once the sensor reads again.

The pass in progress is ended at the last good position, so the map shows a gap over the ground crossed while the sensor was down rather than filling it in with a yield that was never measured. That ground is not counted in the job's acres or bushels.

Manual pause and resume

Press **Pause** on the main screen to manually pause recording. Press **Start** to resume.

Stopping a job

Press **Stop** → confirm. The job is saved and closed. All totals are written to the database. The job remains in the list and can be viewed in the Job Report.

Resume on start

If **Resume Job on Start** is enabled in Settings, the active job is automatically reloaded when YieldFlo restarts. Useful if the PC is rebooted during a harvest day.

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4. Crops

Menu: Menu → Crops

Crops store the grain-specific parameters used in yield calculations and reporting.

Crop settings

Field	Description
Name	Crop name (e.g. Wheat, Corn, Canola)
Category	Grain type category
Bushel Wt	Shown when units are **bu/ac**. The fixed bushel weight for the crop in lb/bu — wheat 60, barley 48, oats 34. Converts harvested mass to bushels.
Test Wt	Shown when units are **t/ha**. The same setting in kg/hL, as quoted on a grain receipt.
Market Moisture	Standard moisture for yield reporting (e.g. 14.5% for wheat)

Adding a crop

1. Press **New** and enter a crop name
2. Set the test weight and market moisture
3. Press **Save**

Select a crop from the list to edit it. At least one crop must exist at all times.

Select a crop and press **Delete** to remove it — you will be asked to confirm. A crop used by a saved job cannot be deleted.

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5. Headers

Menu: Menu → Headers

Headers define the cutting width of the front attachment. Width is used to calculate the area harvested per data point and to draw swath polygons on the yield map.

Header settings

Field	Description
Name	Header name (e.g. 30ft Draper, 8-row Corn Head)
Type	Header type category
Width	Cutting width in feet or metres, following your **Units** setting
Ahead of Pivot	Distance the header sits ahead of the position AOG broadcasts. Enter AOG's **pivot-to-header** distance (from the AOG implement setup). This shifts the recorded coverage to the header, so pass boundaries on the yield map land where the header actually crossed them.

Adding a header

1. Press **New** and enter a header name
2. Enter the cutting width
3. Enter the pivot-to-header distance from your AOG implement setup
4. Press **Save**

At least one header must exist at all times. Changes to a header take effect when a job is started or loaded.

Select a header and press **Delete** to remove it — you will be asked to confirm. A header used by a saved job cannot be deleted.

Note — coverage painted slightly before AOG's: AOG turns its section bits on *early* by its turn-on look-ahead (anticipating valve opening time) but paints its own coverage only where product actually applies. YieldFlo records from the section bits, so each pass start on the yield map can begin a metre or two before AOG's painted coverage. Pass *ends* match exactly. This is normal and harmless for harvest — there is no valve delay on a header — and can be removed in testing by setting AOG's section look-ahead to zero.

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6. Profiles

Menu: Menu → Profiles

Profiles store combine-specific and calibration settings. Use a separate profile for each combine if running YieldFlo on multiple machines.

Profile settings

Field	Description
Name	Profile name (e.g. JD 9600, Case 2388)
Combine ID	Identifier for the module — must match the module configuration
Moisture offset	Fixed offset applied to all moisture readings for this profile
Moisture scale	%/count scale factor for the moisture sensor
Temperature offset	Temperature offset (°C)
Temperature scale	°C/count scale factor for the temperature sensor

> **Note:** Moisture and temperature calibration values are set in **Menu** → **Moisture Cal** and saved automatically to the active profile.

Adding a profile

1. Press **New** and enter a profile name
2. Enter the Combine ID to match your module
3. Press **Save**

At least one profile must exist at all times.

Select a profile and press **Delete** to remove it — you will be asked to confirm. A profile used by a saved job cannot be deleted.

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7. Fields

Menu: Menu → Fields

Fields are optional location labels that can be assigned to jobs for organisation and reporting.

Adding a field

1. Press **New** and enter a field name
2. Press **Save**

Fields are not required. A job can be created without a field selected.

Select a field and press **Delete** to remove it — you will be asked to confirm. A field used by a saved job cannot be deleted.

Importing fields from AgOpenGPS

Press **Import** to open a checklist of field names found in AgOpenGPS's and TWOL's field folders (`Documents\AgOpenGPS\Fields` and `Documents\TWOL\Fields`). Fields already present in YieldFlo are shown but cannot be re-checked. Tick the fields to bring in (or press **Select All**), then press **Import** to add them as new field records — only the field name is imported, not boundary data.

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8. Yield Calibration

Menu: Menu → Yield Cal

Yield calibration corrects the elevator sensor reading to match the actual mass of grain harvested. Two parameters are set here:

- **Sensor Baseline** — the obstruction reading with paddles running but no grain. It belongs to the machine, not the crop: it is stored on the active **profile** and stays put when you change crops.
- **Yield Factor** — a multiplier that scales the sensor reading to match a weighed reference

Setting the baseline

1. Start the clean-grain elevator with no grain
2. Open **Menu** → **Yield Cal**
3. Press **Set Baseline** — it samples for 5 seconds and enters the result in the **Sensor Baseline** field
4. Press **Save**

Nothing takes effect, and nothing is saved, until **Save** is pressed — this applies to both Set Baseline and Calculate.

> **Tip:** Run the empty elevator for at least 10 seconds before setting the baseline so the reading stabilises.

A baseline above 0.25 turns the **Sensor Baseline** field orange, and **Save** asks for confirmation. An empty elevator should read well below this — check for grain or dirt in the elevator and run **Set Baseline** again. You can save the high value if you know it is correct.

Noise readout

Next to the **Set Baseline** button, the **Noise** readout shows how many electrical glitches per second the module is rejecting on the optical sensor signal, averaged over the last 5 seconds. On a healthy installation it reads **0**; the value turns orange when glitches are being rejected, and shows **--** when no module is connected.

A rising noise value — especially one that climbs with elevator speed — indicates electrical interference on the sensor wire (static discharge from the elevator, a chafed or loose wire shaken by vibration, or pickup from nearby wiring). The module filters these glitches out of the yield reading automatically, but persistent noise is worth fixing at the source: check the sensor wire's routing and shielding, connector condition, and the ground path from the sensor bracket and elevator housing to the chassis. The readout works in both sensor signal modes, including Main only.

On module firmware that supports it, a second figure appears next to Noise: **G:n/s**, the rate at which the module is discarding false paddle signals caused by grain crossing the beam between paddles. Unlike Noise this is not an electrical fault — a low rate that rises with flow is normal and needs no action. It shows on both WiFi and CAN connections, and is hidden when the module firmware predates the feature.

Paddle rate readout

Below the **Set Baseline** button, the **Paddles** readout shows how many elevator paddles per second the sensor is seeing, averaged over the last 5 seconds. It shows **--** when no module is connected or the module firmware predates the feature.

Use it to verify the sensor is catching every paddle: the value should match the paddle rate calculated from elevator speed and paddle spacing, and should scale directly with elevator rpm. A reading at half the expected rate means the sensor is missing paddles (alignment or sensing-range problem). On combines with another optical yield monitor sharing the sensor, the value can be compared directly against that monitor's own sensor calibration result — these are usually reported on the monitor's sensor calibration screen as an obstruction percentage at a given frequency in Hz.

Processing delay

The processing delay accounts for the travel time of grain from the header to the elevator sensor. Set this value (in seconds) to match your combine. Typical range is 4–12 seconds.

Two ways to calibrate

The lower half of the Yield Cal screen has two tabs. Both work out a new Yield Factor the same way — comparing what the sensor measured against a real weight — and differ only in where the measured figure comes from.

Tab	Measured figure	Use when
Calibration Run	A pass you start and stop by hand	You can weigh one cart or tank on its own
Whole Field Calibration	The active job's running total	Grain has crossed a scale and you know the total for the job

Weight units

In imperial, the button beside the weight box switches entry between **bu** and **lbs** — press it to change. Enter whichever unit the ticket is in; there is no need to convert by hand.

The choice applies to both tabs and is remembered between sessions. Switching converts whatever is already in the box, so the amount itself does not change. In metric the button reads **kg** and is inactive.

Running a calibration pass

A calibration pass measures the actual mass of grain harvested during a known run, then adjusts the Yield Factor to match.

Preparation: Position a weigh wagon or grain cart to catch grain from the unloading auger and record start and end weights.

1. Open **Menu** → **Yield Cal** and select the **Calibration Run** tab
2. Press **Start Run** — the app begins accumulating sensor data
3. Harvest a suitable area (at least one full tank is recommended)
4. Press **Stop Run**
5. Weigh the harvested grain
6. Enter the actual weight in the **Actual weight** field
7. Press **Calculate**

YieldFlo calculates a new Yield Factor and enters it in the **Yield Factor** field. Press **Save** to store it against the active profile.

Below the **Save** button, a **Baseline last saved** line shows when this profile's Sensor Baseline was last changed. Saving a Yield Factor on its own does not move it — the date answers how long ago the sensor was zeroed. It shows **--** until a baseline has been set.

Entering the weight later

Stop Run freezes the measured total. Harvesting after that does not add to it, so you can carry on combining and enter the weight whenever the grain is next weighed — later the same day, or days later.

The **Measured** line carries the date and time the run was stopped, so a total left standing can be told apart from a fresh one.

While a run is waiting for its weight:

- Pressing **Start Run** again discards it. You are asked first, and the prompt shows the standing total and its date.
- The run is saved, so it survives closing YieldFlo. If the app was closed while a run was still going, it comes back marked **(interrupted)** in orange — grain harvested after the time shown is missing from its total, so a factor fitted to it reads high. **Calculate** warns before using it.
- Changing crop or profile before the weight is entered blocks **Calculate**. The run was measured under one crop, and that is the crop its factor belongs to. Switch back to apply it.

Once the factor has been saved with **Save** the run is spent, and is cleared — the **Measured** line returns to blank. The Yield Factor stays as saved.

Whole Field Calibration

Instead of a dedicated pass, this fits the Yield Factor to the total the active job has already recorded. There is no separate procedure to run: the weight comes from scale tickets you have anyway.

It is not only an end-of-field operation. The job total accumulates continuously, so it can be applied whenever grain has been weighed — after the first few loads, part way through, or when the field is finished.

1. Open **Menu** → **Yield Cal** and select the **Whole Field Calibration** tab
2. Check the **Job** line — it names the job and the crop it was started under
3. Enter the running total in **Total harvested**
4. Press **Calculate**, then **Save**

> **Enter the cumulative total** — everything hauled off this job so far, not just the latest load. A single load fits a factor several times too small, and nothing in the result would reveal it.

After saving, YieldFlo offers to recalculate the job. Accepting rewrites the job's recorded points and total using the new factor, so the map and the job total agree with the weight entered. Declining leaves the job as recorded and changes only the factor used from here on.

Applying more than once is normal. Each recalculation puts the whole job onto the factor being saved, so a later, larger ticket simply supersedes the earlier fit.

Check the yield figure before saving. As the weight is typed, the tab shows the yield it implies over the acres the job recorded, alongside what was actually recorded. If the implied figure is not one the field could have grown,

suspect the acres: ground the app did not record — an auto-pause, or a sensor fault — still crossed the scale, so the recorded total reads short and the factor comes out high. The figure turns orange when it exceeds the recorded yield by half.

The tab is inactive when no job is recording, and says so if the crop has been changed since the job started.

Manual factor adjustment

The Yield Factor can also be adjusted manually in the **Yield Factor** field. Increase to raise yield readings; decrease to lower them.

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9. Moisture Calibration

Menu: Menu → Moisture Cal

Moisture and temperature readings are both calculated the same way:

```
Value = raw_count × scale  +  offset
```

Both support single-point calibration. With offset set to 0, adjust scale until the displayed reading matches a reference instrument:

```
scale = known_value / raw_count
```

For example: if the raw count is 420 and a certified meter reads 18.5%, set scale to 18.5 ÷ 420 ≈ 0.044. The offset field can then be used for fine trimming against a second reference point if needed.

Quick calibration — Calculate

1. Take a grain sample and measure it with a certified grain moisture meter
2. Open **Menu** → **Moisture Cal**
3. With the module connected and grain flowing, note the **App reads:** value
4. Enter the meter's reading in **Meter:**
5. Press **Calculate** — the **Offset** field is filled in for you
6. Press **Save**

Temperature works the same way: enter a thermometer reading in the Temperature section's **Meter:** field and press its **Calculate**.

Full calibration — scale

Calculate shifts every reading by the same amount. If the reading is correct at one moisture but wrong at another, set the scale instead:

1. Set **Offset** to 0
2. With grain flowing, adjust **%/count** until the displayed reading matches the meter
3. Press **Save**

Moisture calibration fields

Field	Saved to	Description
%	Active crop	Moisture offset — set to 0 for single-point scale calibration
%/count	Active profile	Scale factor converting raw ADS1115 counts to moisture percent

Temperature calibration

Same principle as moisture. The default scale of 0.0125 assumes an LM35 sensor (10 mV/°C at PGA = 4.096 V). With offset at 0, adjust **°C/count** until the reading matches a thermometer.

Field	Saved to	Description
°C	Active profile	Temperature offset — set to 0 for single-point scale calibration
°C/count	Active profile	Scale factor converting raw ADS1115 counts to degrees Celsius

> **Note:** Temperature calibration is optional. Temperature does not appear directly on the main screen.

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10. Yield Map

Menu: Menu → Yield Map

The yield map displays a colour-coded spatial plot of yield rate across the harvested area. Each data point is drawn as a filled swath polygon sized to the header width and oriented to the GPS heading at that point.

Colour scale

Yield is mapped to a five-colour gradient:

Colour	Yield
Blue	Lowest
Cyan	Low
Green	Mid
Yellow	High
Red	Highest

The legend at the bottom of the full-screen map shows the low and high yield values for the current data set.

Mini mode

When opened from the menu, the yield map starts as a small floating 300×300 overlay (mini mode). The title bar shows the active job name and zoom controls.

Control	Action
— / **+**	Zoom out / in
×	Close the map overlay
Click the map	Expand to full-screen mode
Drag the title bar	Move the overlay

The mini map position is remembered between sessions.

Full-screen mode

Click the map to expand to full screen. The toolbar at the top provides:

Control	Action
Job selector (drop-down)	Choose which job to display
— / +	Zoom out / in
Recalculate	Re-derives every point's yield for the selected job from its stored raw sensor reading using the crop's *current* calibration, then repaints the map and updates the job total — see below
Print	Export the map as a PNG image — see below
Close	Close the map

Click the map again to return to mini mode. The map can be dragged in both mini and full-screen modes.

Live update during harvest

When the yield map is open and a job is actively recording, the map updates automatically every 2 seconds. New swath polygons are added without re-centering the view.

Recalculating yield

If you change a crop's calibration (Sensor Baseline or Yield Factor) *after* a job was recorded, the job's existing points still reflect the old calibration. Press **Recalculate** in the full-screen toolbar, then confirm the prompt, to re-derive every point's yield from its stored raw sensor reading using the crop's current calibration and repaint the map with the corrected values. The job's total in the Job Report is updated to match. This only affects the selected job and does not change the raw sensor data.

Exporting the map as a PNG

Press **Print** in the full-screen toolbar to export the current map view (including the legend) as a PNG image.

1. A save dialog opens with a default filename of `JobName_Date.png`
2. Choose a folder and confirm
3. The image is saved and a confirmation appears in the status bar

The export folder is remembered for subsequent exports.

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11. Job Report

Menu: Menu → Reports

The job report shows a summary of a completed or active job. Select a job from the list on the left to view its details.

Report contents

Field	Description
Job name	Name entered when the job was created
Field	Field name assigned to the job (if set)
Crop	Crop assigned to the job (if set)
Area	Total area harvested (ac or ha)
Total	Total mass harvested (bu or tonnes)
Avg Yield	Average yield rate across all data points
Avg Moisture	Average grain moisture across all data points
Data Points	Number of recorded GPS data points

Printing a report

Press **Print** to send the job summary to the system print dialog. The report is formatted as plain text suitable for any printer.

CSV export

Press **Export CSV** to save all raw data points for the selected job.

1. A save dialog opens with a default filename of `JobName_Date.csv`
2. Choose a folder and confirm
3. The file is saved and a confirmation appears in the status bar

The export folder is remembered for subsequent exports. The CSV format is compatible with the AgOpenGPS Rate Controller yield overlay.

Column	Description
Timestamp	Date and time of the reading
Latitude	GPS latitude (decimal degrees)
Longitude	GPS longitude (decimal degrees)
WidthMeters	Header cutting width (metres)
Yield_kgha	Yield rate in kg/ha
ElevationMeters	GPS elevation (metres)
Speed_kmh	Ground speed (km/h)
Heading	GPS heading (degrees)
Moisture_pct	Grain moisture (%)
HaAccumulated	Cumulative area at this point (ha)
Sensor1Raw	Raw elevator sensor ratio

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12. Settings

Menu: Menu → Settings

Units

Units setting	bu/ac	t/ha
Yield rate	bu/ac	t/ha
Area	acres	hectares
Speed	mph	km/h
Mass	bushels	tonnes
Header width	feet	metres
Test weight	lb/bu	kg/hL

Changing units takes effect immediately. Historical data is stored internally in acres and bushels and converted for display.

Resume Job on Start

When enabled, YieldFlo automatically reloads the active job when the app starts. Useful if the PC is rebooted during a harvest day.

Saving settings

Press **Save** to store all settings and put them into effect immediately.

Module communication is on its own page — see Comm.

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13. Comm

Menu: Menu → Comm

Sets how the module talks to the app. There is a tab for each connection type: **WiFi / Ethernet** and **CAN**. Tick **Enabled** on the tab you use — only one can be enabled. Press **Save** to apply. Changing the connection type, CAN driver or COM port asks to restart the app.

Each tab shows **Module: Connected** (green) when module data is arriving, **Not Connected** (red) when it is not, or **Not in use** on the tab that is not enabled.

If something is stopping module data from reaching the app, the page tells you when it opens — answer the question, or follow the steps it shows.

WiFi / Ethernet tab

Item	Description
Enabled	The module sends its data over WiFi or wired Ethernet.
Module	Connection status, with the module's IP address when connected.
Set Up Module WiFi	Sends the network name and password to a module. See Setting up a module's WiFi.
Tablet Hotspot	**On** (default): the tablet runs its own WiFi hotspot and the module joins it. **Off**: the module and tablet share an existing WiFi network instead.
Hotspot status	The hotspot's name and state. Red text explains a problem and what to do.

Tablet hotspot

With **Tablet Hotspot** on, the app starts the Windows Mobile Hotspot on 2.4 GHz and keeps it on while the app runs. The module joins it like any WiFi network. Nothing depends on a phone or router being nearby.

The tablet can still get its internet (for GPS corrections) from Ethernet or from WiFi — the hotspot shares it.

The first time, the app may ask:

- **Windows Firewall is blocking module data** — press **Yes**, then **Yes** again in the Windows prompt.
- **A module is in range but not sending data** — press **Yes** to set up its WiFi.

Setting up a module's WiFi

1. Power the module and press **Set Up Module WiFi**.
2. Select the module in the list. Press **Scan** if it is not there.
3. Without the tablet hotspot, enter the password of the WiFi network the PC is on. With the hotspot, nothing needs typing.
4. Press **Set Up Module** and confirm.

Setup takes about a minute. The PC's WiFi leaves its network during that time, so do it when not harvesting — it cannot run while a job is recording. It is finished when the module's data arrives; if it fails, the message says why.

PC WiFi on 5 GHz

The module only works on 2.4 GHz. If the tablet gets its internet from a WiFi network on 5 GHz, the hotspot may not be able to run on 2.4 GHz, and the Comm page says so in red. Set the phone or router to 2.4 GHz if it has that setting, or set the tablet:

1. Press **Windows+R**, type `ncpa.cpl` and press **Enter**.
2. Right-click **Wi-Fi** and choose **Properties**, then **Configure**, then the **Advanced** tab.
3. Set **Preferred Band** to **Prefer 2.4GHz band**. On some PCs it is called **Band** or **Wireless Mode**.
4. Press **OK** and wait for the PC to reconnect.

Wired Ethernet

The module can also send its data over **wired Ethernet** (W5500 board attached to the module). Leave **WiFi / Ethernet** enabled, select Ethernet mode in the module's web portal, and give the PC's wired adapter a static IP on the module's subnet (default `192.168.1.x`).

> **Module setup page:** the module's own settings (communication mode, sensor signals, WiFi credentials, firmware update) are configured on its built-in web page at `http://192.168.200.1`, reached while connected to the module's WiFi hotspot — see Module Web Portal.

CAN tab

Item	Description
Enabled	The module sends its data over CAN bus through a USB CAN adapter.
CAN Driver	The adapter's driver (SLCAN, InnoMaker, or PCAN).
COM Port	Serial/USB port for the adapter (SLCAN driver only). The button beside it refreshes the list.
Adapter	**Connected** / **Not Connected** — for the saved driver and port.

Supported CAN adapters

Driver	Adapters	Notes
SLCAN	CANable (slcan firmware), SH-C30A, other SLCAN adapters	Appears as a COM port — select it under **COM Port**.
InnoMaker	InnoMaker USB2CAN	Requires the vendor's driver and SDK files: copy `InnoMakerUsb2CanLib.dll` and `LibUsbDotNet.dll` from the InnoMaker USB2CAN C# SDK (`Lib` folder) into the YieldFlo application folder. Download them as raw binaries, not from a GitHub web page. The COM Port setting is ignored — the first adapter found is used.
PCAN	Peak PCAN-USB	Install the PEAK device driver package (includes PCAN-Basic) from peak-system.com — no files need to be copied. The COM Port setting is ignored — the first adapter found on PCAN USB channels 1–8 is used.

If CAN fails to start, the reason is written to the error log in `Documents\YieldFlo\Logs`.

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14. Language

Menu: Menu → Language

YieldFlo supports the following languages:

Code	Language
en	English
de	German
fr	French
hu	Hungarian
nl	Dutch
pl	Polish
ru	Russian
lt	Lithuanian

Changing language

1. Press **Menu → Language**
2. Select the desired language from the list
3. Press **Save & Restart**

YieldFlo will restart automatically with the new language applied.

> **Note:** If AgOpenGPS is already installed, YieldFlo will automatically use the same language on first launch.

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15. Module Web Portal

The YieldFlo module has a built-in settings page served from its own WiFi hotspot. Use it to configure the communication mode, the optical sensor signals, and WiFi credentials, and to update the module firmware.

Opening the portal

1. Power the module. It broadcasts a WiFi hotspot named `YieldFlo_ESP32_XXXXXXXX` (the suffix is unique to each module).
2. Connect the PC, tablet, or phone to that hotspot.
3. Browse to `http://192.168.200.1` (with the default module ID 0 — the address is 192.168. (200 + module ID).1).

Settings

Setting	Description
Communication — Mode	WiFi (UDP), CAN bus, or Ethernet (UDP over a wired W5500 board). WiFi/Ethernet and CAN must match the tab enabled on the app's Comm page — the app treats WiFi and Ethernet identically.
Communication — Ethernet subnet	First three octets of the wired network (default `192.168.1`). The module takes IP `subnet.(50 + module ID)`; give the PC's wired adapter a static IP on the same subnet (e.g. `192.168.1.10`). In Ethernet mode the portal shows whether the W5500 board and cable link are detected.
Optical Sensor — Signals	**Main + Comp** (default): both receiver outputs are wired to the module. The module compares them on every edge and rejects electrical noise glitches. **Main only**: only the main signal wire is connected — for example, when sharing the elevator sensor with another yield monitor through a harness that does not carry the complementary wire. Noise rejection is disabled in this mode.
Optical Sensor — Polarity	**PNP** (default — output HIGH with beam clear, the common choice for this kind of sensor) or **NPN** (inverted logic). Select NPN if flow reads high with no grain and low with grain.

Press **Save / Restart** to store the settings in the module and restart it.

WiFi settings are on their own page — follow the **WiFi Network** link at the bottom of the portal.

WiFi network

The **WiFi Network** page joins the module to an existing network and sets the password for the module's own hotspot. The module keeps its hotspot running whether or not it joins a network, so the portal stays reachable either way.

The top of the page shows the current connection.

To join a network:

1. Press **Scan for Networks**. The list appears after a few seconds, strongest first, with a padlock on networks that need a password.
2. Tap a network name to fill it in.
3. Enter the password.
4. Tick **Use this Network**.
5. Press **Save / Restart**.

Setting	Description
Network	Name of the network to join. Tap a scan result to fill this in, or type it in for a hidden network.
Password	Password for that network.
Use this Network	Joins the network. Stays on until you turn it off.
Hotspot — Password	Password for the module's own hotspot. Use 8–10 characters, or leave empty for an open hotspot.

If the module cannot join, it keeps trying in the background and the page shows the reason. **Password refused** means the password is wrong — correct it and press **Save / Restart**. Press **Retry Connection Now** to try again straight away after changing something at the router.

Firmware update

The **Update Firmware** link at the bottom of the portal opens the over-the-air update page. Select the compiled firmware file (`.bin`) and upload it — the module flashes itself and restarts. The installed firmware version is shown at the top of the portal page.

> **Note:** A firmware update that changes the stored settings layout resets the module to factory defaults. Reopen the portal afterwards and re-enter your settings.

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16. Troubleshooting

GPS indicator stays red

- Confirm AgOpenGPS is running on the same PC or local network
- Check that both apps are on the same subnet

Module indicator stays off

- Open **Menu** → **Comm** — it shows the module status and explains common problems when it opens
- Check the module is powered and the WiFi or CAN connection is active
- In WiFi mode, confirm the PC is connected to the correct network
- Check the module LED for status indication
- Restart the module and wait 10–15 seconds for it to connect

Module will not join the tablet hotspot

- On **Menu** → **Comm**, check the hotspot status reads **On**
- If the status says the PC is on 5 GHz, follow PC WiFi on 5 GHz
- If asked about the firewall, press **Yes** and allow the Windows prompt
- Press **Set Up Module WiFi** to send the hotspot details to the module again — needed after a firmware update that resets the module

Module will not join a WiFi network

- Open the module portal and follow the **WiFi Network** link — the page shows the current state
- **Password refused** means the password is wrong; correct it and press **Save / Restart**
- Press **Scan for Networks** to confirm the network is in range
- Press **Retry Connection Now** after changing anything at the router

Yield reads zero or very low

- Check the elevator optical sensor alignment and wiring
- Verify the Sensor Baseline is set correctly (set with elevator running empty)
- Confirm the processing delay is appropriate for your combine
- If the sensor's complementary wire is not connected, set **Signals** to **Main only** in the module web portal — in Main + Comp mode a missing complementary signal causes erratic, low readings

Moisture reads incorrectly

- Verify the moisture sensor is installed and wired correctly
- Check that the moisture calibration (offset and scale) is set in the active profile via **Menu** → **Moisture Cal**
- Compare against a certified grain moisture meter and adjust the calibration

App crashes on start

- Check `YieldFlo_Crash.log` in the application folder for details

Job does not resume after restart

- Ensure **Resume Job on Start** is enabled in Settings
- The job must have been active (not stopped) when YieldFlo was closed

AOG UDP failed to start

- If AgOpenGPS Rate Controller (RC) is running, both RC and YieldFlo must be built from current source for UDP port sharing to work
- YieldFlo will still function for module data — only GPS reception will be affected

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YieldFlo is designed for use with AgOpenGPS. It is not a certified weighing system and should not be used as the sole basis for grain settlement.